Branch-specific detection of phylogenetic signal with SatuTe

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Abstract

Sequence saturation can erase the historical signal for a branch despite high conventional support. SatuTe tests whether the subtrees separated by a branch remain statistically coherent. We extend its spectral statistic to all non-stationary directions of a reversible substitution model, weighted by their persistence, and derive the correct null for invariant-site mixtures. The extension reduces exactly to published SatuTe under Jukes–Cantor. In paired publication-scale simulations, weighting improved detection chiefly for short alignments at the long-branch decision boundary; the largest matched GTR+F+G4 increase was 57.6 percentage points under taxon Bonferroni. Across 1,728 matched-protein cells, differences were positive in 831 and negative in none. Independent truth-labelled experiments verified false-discovery control. Tree-of-Life and HCV examples localized broad rRNA and codon-position-specific losses, respectively. SatuTe therefore complements branch support by identifying where alignments retain recoverable historical signal.

Model-based phylogenetics combines tree search with branch-support measures such as the bootstrap and approximate likelihood-ratio tests [1, 2, 3, 4]. These measures quantify reproducibility or local preference for a split. They do not test whether repeated substitutions have erased the historical dependence connecting the groups on its two sides. A branch can therefore have high support even when the alignment no longer contains recoverable signal across that branch.

Saturation is the progressive loss of historical information as multiple substitutions accumulate at the same sites [5, 6]. Alignment-wide indices and locus filters can identify heavily saturated data [7, 8], but deep phylogenies often contain a mixture of informative and uninformative genes, regions and rate categories. A branch-specific diagnostic is needed to localize that loss.

SatuTe (Saturation Test) generalizes saturation from pairs of sequences to the two subtrees separated by a branch [9]. Its null hypothesis is subtree independence, represented by independently evolving subalignments. Rejection indicates phylogenetic signal and classifies the branch as phylogenetically informative; otherwise, the branch is considered saturated at the chosen threshold. SatuTe thus complements branch support: one asks whether a split is reproducible, the other whether the alignment still connects its two sides.

The test uses the spectral decomposition of a reversible substitution model [10, 11]. For a site pattern ∂, partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$ give subtree probabilities $P(\partial_A) = \langle L(\partial_A), \pi \rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi \rangle$. With eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots$ and left eigenvectors h_i, the coherence coefficient is

$$C_{\partial i} = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle . \tag{1}$$

Published SatuTe retains only the slowest non-stationary component. We ask whether faster components should contribute while they persist on a finite branch, and derive weights from their decay relative to the slowest component. We also correct the infinite-branch null for explicit invariant sites, which retain ancestral dependence even as the focal branch tends to infinity.

We benchmark both changes against the original Jukes–Cantor (JC) design and in paired nucleotide and protein simulations with three decision rules and independent truth-labelled

calibration. We also revisit ribosomal-protein two-domain and 16S rRNA three-domain Tree-of-Life examples [12, 13, 14]. The aim is not to choose a Tree-of-Life topology, but to identify which branches and alignment regions retain detectable historical signal under a fitted model.

Results

We evaluated the extension in a prespecified sequence. We first derived the persistence-weighted statistic and its invariant-mixture null, then verified exact agreement with published SatuTe under JC and reconstructed the original simulation behavior. We next tested null centering, paired power and model dependence under nucleotide and protein models, followed by independent truth-labelled error-control experiments. Finally, we asked whether the resulting branch- and region-specific diagnoses remained interpretable in Tree-of-Life and viral alignments.

Persistence weighting extends SatuTe without changing its decision

The spectral decomposition used by SatuTe gives

$$\frac{P(\partial \mid t)}{P(\partial_A)P(\partial_B)} = 1 + \sum_{i=1}^{d} C_{\partial i} \exp(\lambda_i t). \tag{2}$$

In Eq. (2), d is the number of non-stationary eigenvalues: $d = 3$ for DNA and $d = 19$ for an amino-acid model. The coefficient $C_{\partial i}$ measures agreement between the two subtrees in spectral direction h_i. The factor $\exp(\lambda_i t)$ measures how much of that direction remains after evolution across branch AB. Because all non-stationary eigenvalues are negative, components with more negative λ_i decay faster.

We define the eigenvalue-weighted coherence for pattern ∂ as

$$C_{\partial}^{\lambda}(t) = \sum_{i=1}^{d} w_i(t) C_{\partial i}, \qquad w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}. \tag{3}$$

The relative weight $w_i(t)$ sets $w_1(t) = 1$ and downweights components that decay faster than the second-largest-eigenvalue component. Thus the published SatuTe coefficient remains the reference scale, while the other components contribute according to the fitted branch length.

For an alignment of n independently evolving sites, the empirical coefficient is

$$\hat{C}_{\lambda}(t) = \frac{1}{n} \sum_{s=1}^{n} C_{\partial_s}^{\lambda}(t). \tag{4}$$

The variance estimator has the same form as the SatuTe estimator for a repeated second largest eigenvalue, with the mode weights included:

$$\hat{\sigma}_{\lambda}^2(t) = \sum_{i=1}^{d} \sum_{j=1}^{d} w_i(t) w_j(t) \hat{\sigma}_{A,i,j}^2 \hat{\sigma}_{B,i,j}^2. \tag{5}$$

The corresponding test statistic is

$$Z_{\lambda}(t) = \frac{\sqrt{n}\hat{C}_{\lambda}(t)}{\hat{\sigma}_{\lambda}(t)}. \tag{6}$$

The decision rule is unchanged. Branch AB is called phylogenetically informative when $Z_{\lambda}(t) > z_{\alpha}$. When the tree is inferred from the same alignment, the same Bonferroni correction used by SatuTe gives the threshold $z_{\alpha/(m_A m_B)}$.

SatuTe in a nutshell

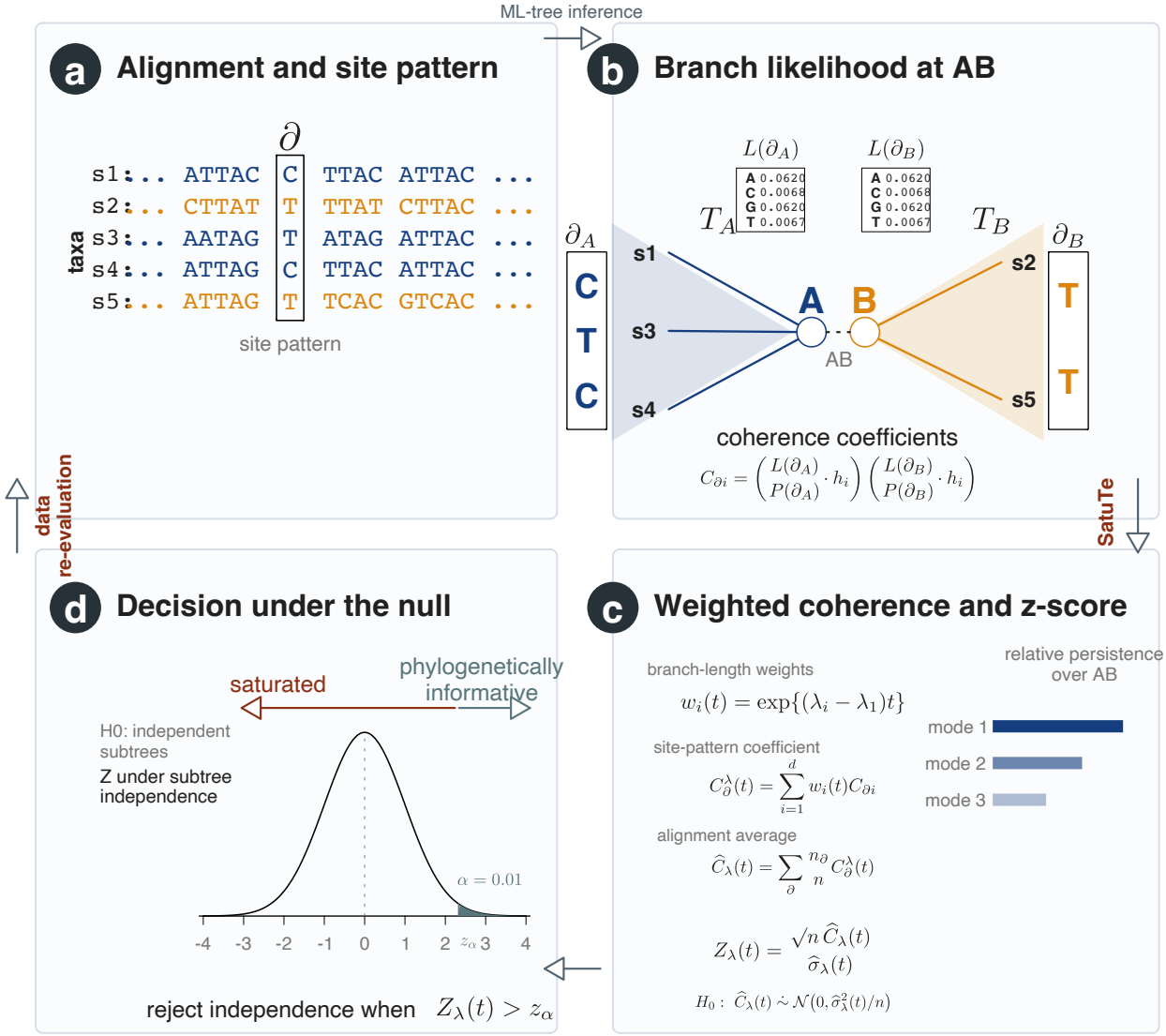


Figure 1: **SatuTe in a nutshell. a,** Columns of a five-taxon alignment define site patterns ∂. **b,** Branch AB partitions each pattern into subpatterns ∂_A and ∂_B; their partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$ define the coherence coefficients. **c,** Non-stationary components are weighted by persistence relative to the slowest component over branch length t, and average coherence is transformed into a z-score. **d,** Rejection of subtree independence indicates phylogenetic signal and classifies the branch as phylogenetically informative; otherwise, the branch is considered saturated at the selected threshold.

Invariant categories require a different null likelihood

An explicit invariant category is a boundary case of the rate mixture. Let q_c and r_c be the proportion and rate of category c, and let A_{sc} and B_{sc} be the two subtree-pattern probabilities for site pattern s. For a positive-rate category, the infinite-branch null contribution is $q_c A_{sc} B_{sc}$. For the invariant category $c = 0$, however, $r_0 = 0$ and the transition matrix remains the identity even as the focal branch length tends to infinity. Its null contribution is therefore the joint probability

$$J_{s0} = \sum_a \pi_a L_{A,s0}(a) L_{B,s0}(a), \tag{7}$$

rather than the product $A_{s0}B_{s0}$. Retaining the joint term in Eq. (7), the correct mixture null is

$$p_\infty(s) = q_0 J_{s0} + \sum_{c:r_c>0} q_c A_{sc} B_{sc}. \tag{8}$$

For the positive-rate categories, define the model-weighted site score

$$u_s(t) = \frac{\displaystyle\sum_{c:r_c>0} q_c A_{sc} B_{sc} \sum_{i=1}^{d} e^{\lambda_i r_c t} C_{sc,i}}{p_\infty(s)}. \tag{9}$$

The invariant component enters only the denominator of Eq. (9) and contributes no finite-versus-infinite-branch signal. This gives the exact identity

$$\frac{p_t(s)}{p_\infty(s)} = 1 + u_s(t). \tag{10}$$

In computation, all positive-rate persistence weights are divided by one common factor e^δ, where $\delta = \max_{c:r_c>0,i} \lambda_i r_c t$. This prevents underflow without changing the studentized statistic. The refined test uses the sample mean and sample variance of these globally rescaled site scores. We refer to it as the refined invariant-null statistic.

The previous counterfactual construction treated the invariant category as if it became independent across an infinite focal branch, replacing J_{s0} by $A_{s0}B_{s0}$ and retaining a zero-rate numerator contribution. That construction does not satisfy Eq. (10) under a $+I$ model and need not have null expectation zero.

The Jukes–Cantor case is unchanged

The Jukes–Cantor (JC) model has one stationary eigenvalue and a three-dimensional non-stationary eigenspace with $\lambda_1 = \lambda_2 = \lambda_3$. Therefore $w_i(t) = 1$ for all non-stationary modes in Eq. (3). The weighted statistic becomes

$$C_\partial^\lambda(t) = \sum_{k=1}^{3} C_{\partial 1,k} = C_{\partial 1}, \tag{11}$$

The collapse in Eq. (11) is the multiplicity-three expression used by SatuTe. Thus the weighted statistic does not change the JC statistic. If the substitution model does not distinguish among non-stationary directions, the weighted statistic does not introduce that distinction.

JC reconstruction recovers the original SatuTe behavior

The exact JC collapse provides a no-change control for the extension. We therefore retained the original SatuTe simulation design: five- and 16-taxon trees, 100–10,000 sites, focal lengths 0.1–10 and 1,000 JC replicates per point. Four analyses separate branch-length estimation from

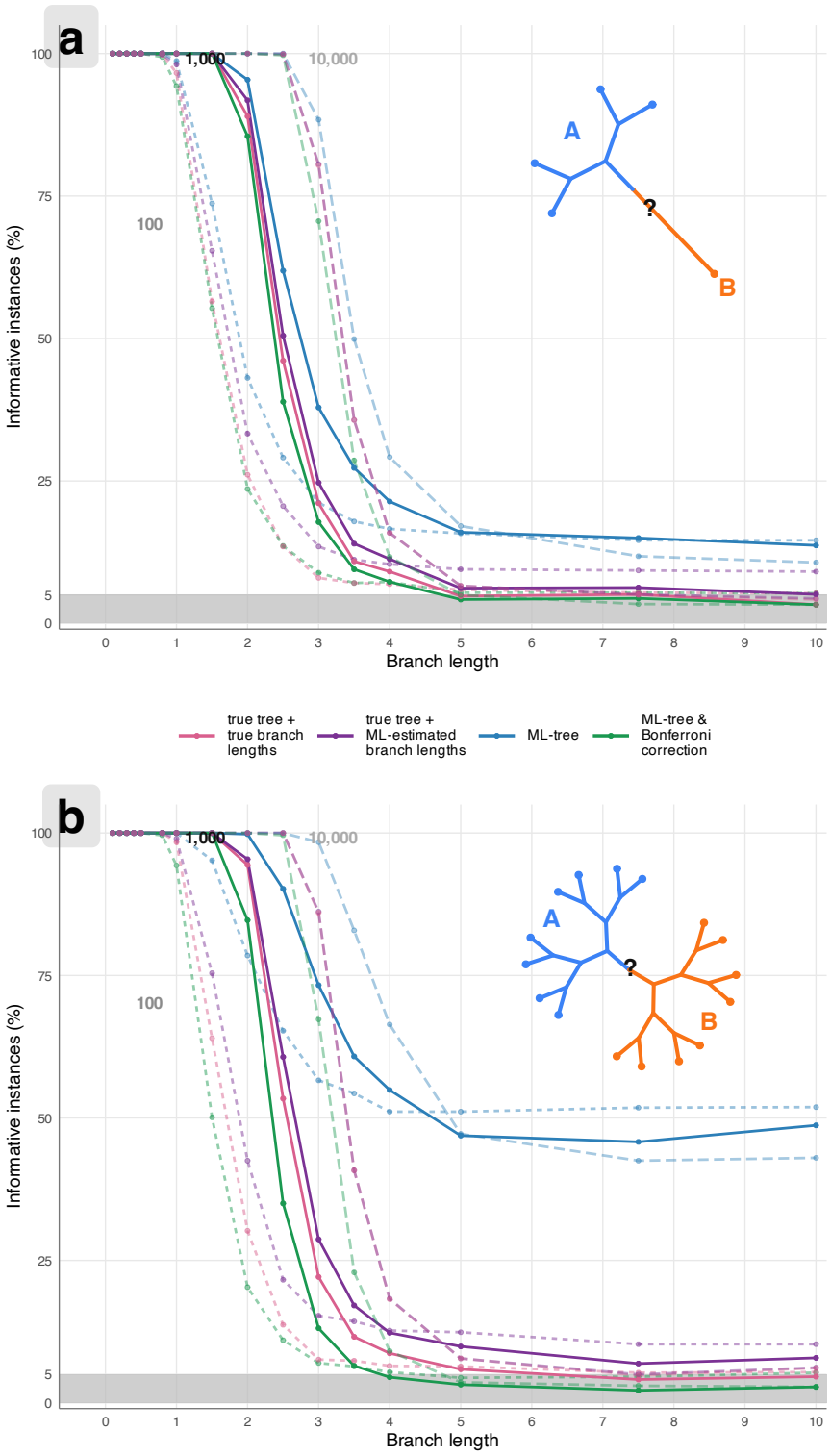


Figure 2: **Direct comparison with the original SatuTe simulation design.** The current implementation repeats the original paper's JC grid for five-taxon external and 16-taxon internal targets. Colors distinguish the true-tree reference, maximum-likelihood branch-length estimation, reuse of the alignment for topology selection and testing, and taxon-pair Bonferroni correction. Line styles show 100, 1,000 and 10,000 sites; every point contains 1,000 replicates. The unrooted tree schematics are rendered from `ape` phylogenies with `ggtree`.

topology selection: true tree and lengths; true tree with maximum-likelihood (ML) lengths; an ML tree and lengths; and the same inferred-tree analysis with taxon-pair Bonferroni correction.

With the true tree and lengths, the fraction of phylogenetically informative alignments fell from near one to the 5% null level as the focal branch lengthened; longer alignments retained signal further (Fig. 2). ML length estimation on the true topology closely followed this reference. Selecting and testing a topology on the same alignment was liberal at long branches, reaching approximately 15% for the external and 50% for the internal target. Taxon-pair Bonferroni correction largely restored the reference curve, but became conservative for subtrees with nearly indistinguishable sequences.

For 100-site 16-taxon alignments, correct topology recovery fell from 63.2% at focal length 0.1 to 4.5% at length 1 and nearly zero beyond length 3. Yet bootstrap support for the split reached 100% by length 0.8 and stayed there while Bonferroni SatuTe calls fell to the 3.5–4.6% null range (Supplementary Tables S2 and S3). This establishes that high split support does not imply recoverable history across the split.

JC is the prespecified formula control: its three non-stationary directions have equal eigenvalues, so weighted and published SatuTe curves coincide exactly. Models with distinct eigenvalues separate chiefly for short alignments at the long-branch detection boundary.

Decision changes are concentrated on the focal branch

Before extending the comparison beyond JC, we tested whether weighting created a nonspecific tree-wide score shift. At branch length 12 and 100 sites, focal-branch increases were 14.5–22.0 percentage points under GTR+G4 and 38.1–39.1 points under LG+G4, whereas adjacent branches changed by essentially zero (Supplementary Fig. S1). Thus the added signal was localized to the deliberately lengthened branch.

The corrected invariant null restores false-positive control

Having established exact continuity with the original JC test, we used exact enumeration to validate the finite-branch $+I$ null. This exploratory $+I + G4$ and $+I + R4$ grid is distinct from the fixed-model publication benchmark below. Across 270 model–branch–alignment-length cells, 10.8 million paired alignments were evaluated. The likelihood identity in Eq. (10) held to 1.55×10^{-11}. The refined score had exact null expectation below 1.53×10^{-16} in absolute value, whereas a counterfactual score that treated invariant sites as independent retained expectations of 0.0354–0.2176.

At the ordinary one-sided $N(0, 1)$ threshold, counterfactual rejection averaged 0.900; refined rejection averaged 0.0492. Split-sample empirical thresholds brought held-out rejection near 0.05 for both formulas, but did not make the counterfactual statistic a standard-normal pivot: at $t = 10{,}000$, its threshold rose from 3.15 at 100 sites to 18.47 at 10,000 sites, whereas the refined threshold remained between 1.62 and 1.69.

The refined invariant statistic gains power at the detection boundary

After independent calibration, the refined statistic had the larger exact null-centered separation in 245 of 270 cells and equal separation in the remaining 25. Gains formed a diagonal ridge as longer alignments moved the detection boundary to longer branches (Fig. 3). The maximum $+I + G4$ gain was 0.116 at $t = 60$ and 10,000 sites (bootstrap 95% interval, 0.111–0.122); the maximum exploratory $+I + R4$ gain was 0.163 (0.157–0.170). Supplementary Table S1 reports maxima by alignment length.

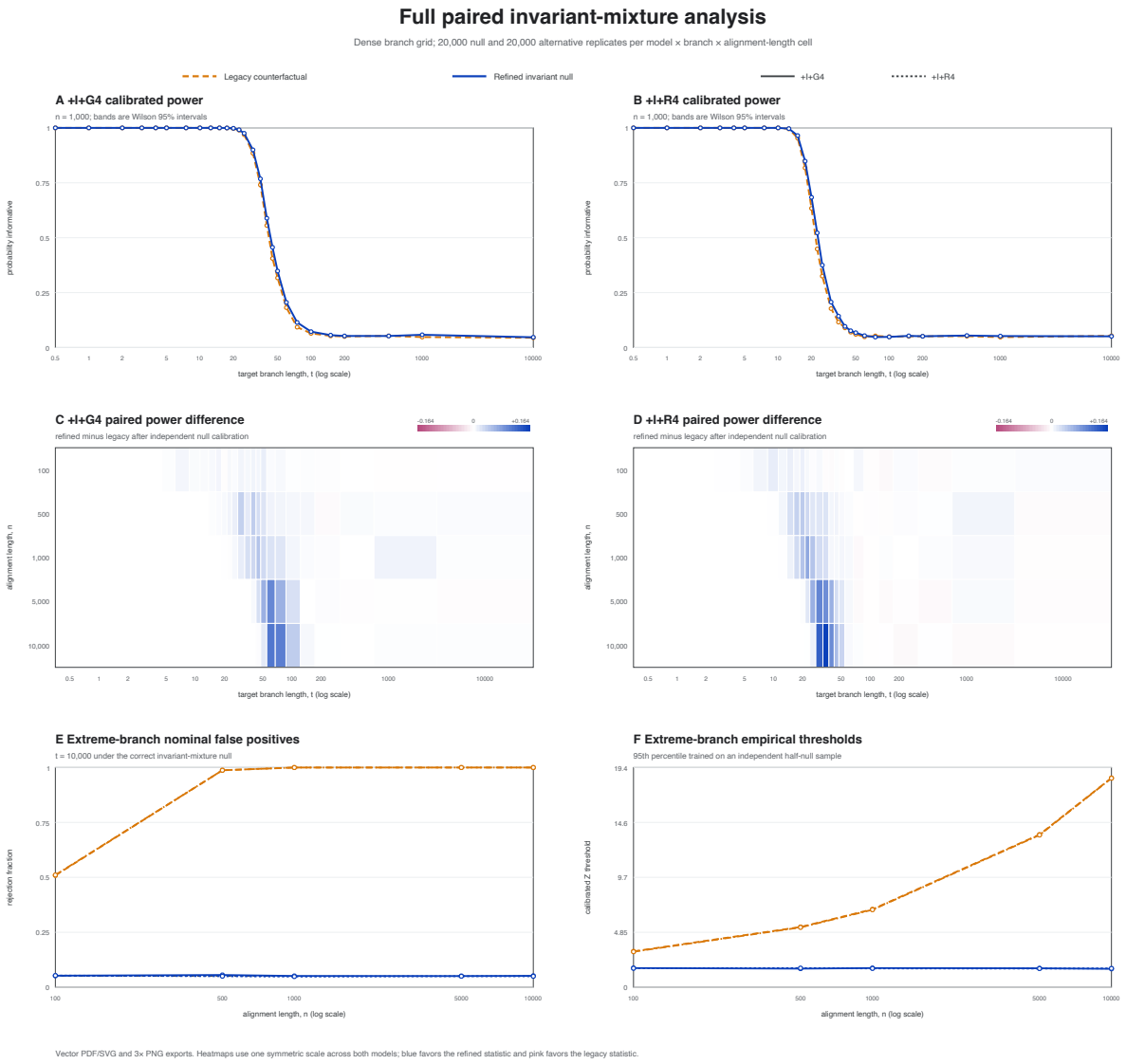


Figure 3: **Correcting the invariant-site null restores calibration. a,b,** Empirically calibrated power at 1,000 sites across the branch grid; bands are Wilson 95% intervals. **c,d,** Refined-minus-counterfactual power for five alignment lengths. Blue favors the refined statistic and pink the counterfactual. **e,** Nominal false-positive rates at $t = 10{,}000$. **f,** Empirical 95th-percentile thresholds trained on an independent half-null sample.

Publication-scale nucleotide power curves

We next tested whether the additional spectral directions improved detection under models that distinguish among them. Fixed GTR+F, GTR+F+G4 and GTR+F+I+G4 models were evaluated in 1,000 paired replicates per cell, with 100, 250 or 1,000 sites; focal lengths 4–12; and five-taxon external or 16-taxon internal targets. The merged grid contains 864,000 detail rows and 2,016 summaries, with complete replicate identifiers and no JC-collapse mismatch.

Both GTR+F+G4 statistics lost power as the focal branch length increased, and longer alignments shifted the transition rightward (Fig. 4). Eigenvalue weighting produced the largest shift at 100 sites, where published SatuTe left the high-power regime while the eigenvalue-weighted SatuTe statistic still detected coherence. Supplementary Figs. S3–S5 provide paired differences and the corresponding GTR+F and GTR+F+I+G4 curves.

The largest GTR+F+G4 increase was 0.241 unadjusted (95% bootstrap interval, 0.215–0.269), 0.414 under Benjamini–Yekutieli (BY; 0.384–0.446) and 0.532 under taxon-pair Bonferroni in the inferred-ML-tree analysis (0.501–0.562). All occurred at 100 sites on the 16-taxon target. Exact McNemar P values were 5.7×10^{-73}, 4.7×10^{-125} and 1.4×10^{-160}, respectively. Taxon-pair Bonferroni is not defined for either fixed-topology analysis and was therefore not applied to those scenarios.

Model misspecification changes absolute operating characteristics

We next simulated GTR+F but evaluated the same alignments under GTR, JC, K2P or F81. Across 960,000 detail rows, JC and K2P were more liberal than correctly specified GTR, whereas F81, which retains base composition, was substantially closer (Fig. 5). Thus a small paired formula difference under a misspecified model does not imply correct absolute calibration.

Within misspecified analyses, the largest weighted gains were 0.131 unadjusted, 0.206 under taxon Bonferroni and 0.209 under BY. JC remained an exact formula-control because its non-stationary eigenvalues are equal. Supplementary Figs. S10–S12 show JC, K2P and F81 paired differences and weighted curves.

Matched protein power varies strongly among models

The matched protein suite used fixed LG+G4, WAG+G4, JTT+G4 and Q.pfam+G4 models. Eigenvalue-weighted-minus-published-SatuTe differences were never negative among the applicable scenario–rule cells. In the inferred-ML-tree analysis, maximum taxon-pair-Bonferroni gains were 0.268 for LG, 0.410 for WAG, 0.546 for JTT and 0.304 for Q.pfam. The consolidated inferred-tree comparison shows the exact JC control beside the GTR+F+G4 and LG+G4 separation (Fig. 6); complete published-SatuTe, eigenvalue-weighted-SatuTe and paired-difference curves show the matrix-specific transitions (Supplementary Figs. S6–S9).

Independent truth-labelled calibration

Independent calibration used 5,000 exact-null replicates per cell, split equally between threshold training and held-out evaluation, plus 2,000 50:50 mixed-null replicates. In the GTR mixed-null design, power ranged from 0.929 to 0.989 and empirical FDR from zero to 0.00386. Formula-paired differences were at most 0.000571 for power and 0.000143 for FDR, confirming that the larger publication-grid effects are localized to the saturation boundary (Fig. 7).

Across 96 cells, held-out rejection ranged from 0.0108 to 0.0644; unadjusted and BY GTR cells ranged from 0.0432 to 0.0596. Sixteen-taxon Bonferroni cells were conservative because only complete tied blocks of discrete adjusted P values were attainable.

The Tree-of-Life examples expose alignment-specific loss of signal

Finally, we examined whether the controlled gains translated into interpretable local diagnoses in biological alignments. We revisited two datasets used for deep Tree-of-Life inference [12]: 2,596 amino-acid sites from 16 ribosomal proteins across 3,083 taxa, supporting a two-domain (2D) tree; and 1,947 16S rRNA sites across 1,871 taxa, supporting a three-domain (3D) tree. Branch lengths were re-estimated under LG+G4 and GTR+G4, respectively. We tested the internal branch leading to Eukaryota and an external yeast branch (Supplementary Fig. S2).

In the protein 2D tree, all 6,163 branches were informative in each of four rate categories. In the fastest rRNA category (rate 2.73), 107 of 3,739 branches were saturated; 13 were internal, including the Eukaryota branch.

Sliding windows localized this contrast (Fig. 8). Published SatuTe classified 822 of 1,912 rRNA windows (43.0%) as saturated for the Eukaryota branch, compared with 138 of 2,561 protein windows (5.4%). Protein losses were short and often coincided with boundaries in the concatenation. rRNA losses were enriched in hypervariable regions V1–V9 [15]; conserved regions C1 and C4–C9 retained signal, whereas C2 and C3 were nearly saturated. Gappy intervals at positions 240–301 and 789–956 contributed little information.

Eigenvalue weighting reduced, but did not erase, the pattern. Native recomputations of the published and eigenvalue-weighted SatuTe statistics classified 895 and 699 rRNA windows as saturated for Eukaryota, and 83 and 29 for yeast. In proteins, the published SatuTe analysis and native eigenvalue-weighted recomputation yielded 138 and 5 saturated windows for Eukaryota, and 54 and 8 for yeast. Because these cross-implementation comparisons also differ in input handling, paired simulations provide the controlled formula-effect estimate.

All 16 ribosomal proteins produced significant scores under both 2D and rearranged 3D placements of Eukaryota; L15, L22 and S10 favored the rearranged placement. Six of nine rRNA regions had larger scores under the 3D placement. The two distributions of score differences were not significantly different (Kolmogorov–Smirnov $p = 0.072$). These results localize signal for candidate placements rather than select a true topology (Supplementary Tables S4 and S5).

An independent hepatitis C virus (HCV) reference panel provided a coding-genome example spanning seven recognized genotypes [16]. Every complete gene and codon-position partition retained signal, but 36-site z-score curves exposed localized losses (Fig. 9). Across genotype 1, 2 and 3 branches, published SatuTe classified 1/507, 4/507 and 5/507 positions-1+2 windows as saturated, compared with 11/252, 34/252 and 33/252 third-position windows. Eigenvalue weighting reduced the corresponding third-position counts to 6, 24 and 27 (Supplementary Table S6). This contrast is informative because a gene-level screen would retain every locus, whereas the curves expose third-position and interval-specific signal loss hidden by aggregation. It motivates partition-specific model checks or masking rather than wholesale removal of otherwise informative genes.

Together, the reference analyses reproduce the scale at which the simulation gains occur. Weighting changes conclusions near a local detection boundary, but does not convert broadly saturated rRNA regions into uniformly informative data or alter the contrast between codon partitions. The practical result is therefore not a general increase in significance: it is a more sensitive branch-specific test whose gains remain tied to identifiable alignment regions.

Discussion

This study establishes a continuous extension of SatuTe rather than a replacement test. Persistence weighting leaves the JC statistic exactly unchanged, adds power only when a fitted model contains distinguishable non-stationary directions, and preserves error control in independent truth-labelled experiments. Correct treatment of invariant sites separately restores the null that a zero-rate category otherwise violates. Across these settings, the extension improved

sensitivity at the long-branch detection boundary without changing the branch, null hypothesis or one-sided decision that define SatuTe.

The historical JC reconstruction clarifies what this decision measures. SatuTe asks whether an alignment retains detectable dependence between the subtrees separated by a branch. Bootstrap and likelihood-ratio support instead quantify reproducibility or preference for a split [1, 2, 17]. In the 16-taxon control, bootstrap support remained complete after topology recovery collapsed and SatuTe approached its null rejection rate. A non-significant SatuTe result therefore identifies insufficient branch-specific signal in the tested alignment and model; it does not deny common ancestry.

The gain was deliberately narrow rather than universal. It appeared chiefly for short alignments crossing the long-branch detection boundary and was negligible when both formulas had high power or both had reached the null floor. Branch-neighbourhood audits showed that weighting did not indiscriminately raise scores across a tree. Its magnitude depended strongly on the substitution matrix: JTT produced broader shifts than LG or Q.pfam, demonstrating that alphabet size alone does not determine the recoverable spectral information.

Model adequacy remains at least as consequential as the choice of statistic. JC and K2P analyses of GTR data were too liberal, whereas F81 was closer to the correctly specified result. The JC zero-difference control establishes formula equality, not robustness of JC to misspecification. Likewise, invariant sites create a distinct structural problem: a zero-rate category remains ancestrally dependent across an infinite focal branch, so placing it in a product-form null introduces a positive mean that studentization amplifies with alignment length. Including its joint probability only in the null denominator restored exact centering and standard-normal calibration. The exploratory $+I + R4$ analysis is a stress test; fixed GTR+F+I+G4 provides the publication benchmark.

The empirical examples show why a local diagnostic is useful after these statistical controls are satisfied. Ribosomal proteins retained signal across nearly all windows, whereas 16S rRNA contained broad saturated regions and lost signal for the Eukaryota branch in its fastest category. In HCV, gene-level tests were uniformly informative, yet short-window losses concentrated at synonymous-enriched third positions. An alignment-wide pass can therefore mask within-locus saturation. Branch- and window-specific curves can instead direct model checks, partitioning or selective masking without discarding an otherwise informative locus. Weighting reduced threshold crossings without erasing either biological contrast.

The interpretation remains bounded. These reference-panel analyses localize signal; they neither select the true topology nor estimate viral population diversity. The HCV result is additionally limited by ten genomes, overlapping short windows and model refitting at that scale. Significance also does not guarantee correct topology or model adequacy. Non-stationary or non-reversible extensions would make the root identifiable and require a root-aware null.

Within the present reversible framework, the evidence supports a clear workflow. Published SatuTe remains the reference statistic and exact JC control; the weighted result reports sensitivity gained from spectral directions that persist under the fitted model; the corrected mixture null is required when invariant sites are explicit. Both statistics should be reported with paired differences, multiplicity correction, conventional branch support and model-adequacy checks. Together, these views connect the theoretical extension, controlled operating characteristics and biological examples: they make loss of historical signal testable where phylogenetic conclusions are actually made, at individual branches and within specific alignment regions.

Online Methods

Notation

We use the notation of SatuTe. A branch AB separates subtrees T_A and T_B. A site pattern ∂ induces subpatterns ∂_A and ∂_B. The partial likelihood vectors at the endpoints are $L(\partial_A)$ and $L(\partial_B)$. The stationary distribution is π, with $P(\partial_A) = \langle L(\partial_A), \pi \rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi \rangle$. The reversible rate matrix has eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ and left eigenvectors h_i, where d is the number of non-stationary components. Coherence coefficients are defined by Eq. (1).

Weighted estimator

For each site s, $C_{\partial s i}$ is computed for the non-stationary modes. For branch length t, the mode weight is $w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$. The quantities $\hat{C}_\lambda(t)$, $\hat{\sigma}^2_\lambda(t)$ and $Z_\lambda(t)$ are then computed using Eqs. (4) to (6). If a model has a repeated second largest eigenvalue, each direction in that eigenspace has weight one. For JC, the estimator is the published SatuTe statistic.

Null distribution

The weighted statistic keeps the SatuTe null hypothesis. Let

$$X_i(\partial_A) = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle, \qquad Y_i(\partial_B) = \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle.$$

Then $C_{\partial i} = X_i Y_i$ and

$$C^\lambda_\partial(t) = \sum_{i=1}^{d} w_i(t) X_i Y_i.$$

Under the SatuTe null hypothesis, the two subtree patterns ∂_A and ∂_B are independent. The non-stationary eigenvectors are orthogonal to the stationary component, so the original SatuTe argument gives $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[Y_i] = 0$. Therefore $\mathbb{E}[C_{\partial i}] = 0$ for every non-stationary component, and by linearity $\mathbb{E}[C^\lambda_\partial(t)] = 0$.

The variance follows by expanding the square of the weighted sum. Define the subtree second moments

$$\sigma^2_{A,i,j} = \mathbb{E}[X_i X_j], \qquad \sigma^2_{B,i,j} = \mathbb{E}[Y_i Y_j].$$

Using independence of the two subtrees under the null,

$$\mathrm{Var}\Big(C^\lambda_\partial(t)\Big) = \sum_{i=1}^{d} \sum_{j=1}^{d} w_i(t) w_j(t) \sigma^2_{A,i,j} \sigma^2_{B,i,j}. \tag{12}$$

Consequently, Eq. (12) implies that, for n independent sites, $\mathrm{Var}[\hat{C}_\lambda(t)] = \mathrm{Var}(C^\lambda_\partial(t))/n$, which is the population form estimated in Eq. (5). With finite fourth moments and a consistent variance estimator, the central limit theorem and Slutsky's theorem give $Z_\lambda(t) \xrightarrow{d} N(0,1)$ under subtree independence. Maximum-likelihood topology selection remains a separate model-selection step, so the same Bonferroni correction used by SatuTe is retained for that scenario.

Simulation protocol

The five-taxon tree is a balanced four-taxon subtree T_A, with all subtree branches set to 0.2, and a single taxon B attached by focal branch AB. The 16-taxon tree contains two balanced eight-taxon subtrees; internal and external lengths are drawn from EvoNAPS-derived distributions

and restricted to 0.04–0.2. The original SatuTe baseline uses JC, $n \in \{100, 1{,}000, 10{,}000\}$, 1,000 replicates per cell and focal lengths

$$t \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.8, 1, 1.5, 2, 2.5, 3, 3.5, 4, 5, 7.5, 10\}.$$

Every alignment is analysed under four scenarios: the true topology and lengths; the true topology with ML lengths; an ML-inferred topology and lengths; and that inferred-tree analysis with threshold $\alpha/(m_A m_B)$. For the five- and 16-taxon targets, the Bonferroni denominators are 4 and 64, respectively. Baseline alignments were simulated with Seq-Gen and evaluated with IQ-TREE and SatuTe [18, 4].

The rate-heterogeneous extensions use full alignments and paired evaluation of the published and eigenvalue-weighted SatuTe statistics. Within a replicate, both statistics use the same alignment, tree, branch length, target split and random seed. This isolates the effect of eigenvalue weighting from simulator and topology variation.

The nucleotide simulations use the Tree-of-Life 16S rRNA GTR+F+G4 setting, implemented as a fixed four-category model in the IQ-TREE integration. Exploratory protein simulations use the Tree-of-Life LG+G4 setting. The growing-site grid uses

$$n \in \{100, 250, 500, 1{,}000, 2{,}000, 4{,}000, 10{,}000\},$$

focal lengths $4, 5, 6, 7.5, 10, 12$ and 100 replicates per cell. The corrected publication suite uses 1,000 replicates, $100, 250, 1{,}000$ sites and focal lengths $4, 5, 6, 7, 8, 9, 10, 12$. It evaluates fixed GTR+F, GTR+F+G4, GTR+F+I+G4, LG+G4, WAG+G4, JTT+G4 and Q.pfam+G4 models. The branch-neighbourhood audit computes both statistics for the target and adjacent branches in the same trees.

For model misspecification, alignments are generated under fixed GTR+F and evaluated under GTR, JC, K2P or F81. This separates equality of the two SatuTe statistics under JC from adequacy of JC for GTR-generated data. Fractions of phylogenetically informative alignments are reported for each statistic in addition to paired differences.

The independent calibration experiment uses 5,000 exact-null replicates in every cell, prespecifying replicates 1–2,500 for threshold training and 2,501–5,000 for held-out evaluation. A separate 2,000-replicate design alternates target-null and target-alternative branches to form a 50:50 mixed-null sample for empirical false-discovery rate and power. For discrete adjusted p-values, the training-derived cutoff is the largest attainable value for which the complete tied block has empirical probability at most 0.05; if the minimum value already exceeds that mass, the cutoff is placed immediately below it. This tie-aware definition makes the fitted cutoff agree with the implemented $p \leq c$ decision rule. We report Wilson intervals for held-out null rejection and paired bootstrap intervals for differences between formulas.

All publication outputs use schema-v2 detail, summary, branch-audit, decision-rule-audit and timing tables. Replicate completeness is checked within every model–tree–scenario–site–branch cell before aggregation. Paired percentile-bootstrap intervals resample replicate-level discordance, and exact McNemar tests use the two discordant call counts.

Paired invariant-mixture grid

The $+I$ analysis uses a four-taxon tree with two two-tip subtrees, terminal branches of length 0.05 and a varied focal branch. The substitution process is fixed to GTR with exchangeabilities $\{1, 2, 1, 1, 2, 1\}$ and stationary frequencies $\{0.30, 0.20, 0.20, 0.30\}$. Both models have invariant proportion $q_0 = 0.10$. The $+I+G4$ model uses four equal-probability discrete-gamma categories with shape 0.5. The $+I+R4$ model uses positive-category proportions $\{0.35, 0.25, 0.23, 0.17\}$ and rates $\{0.0747656, 0.420849, 1.15850, 3.54214\}$, rescaled after addition of the invariant component so that the overall mean rate is one.

The $+I+R4$ construction is retained only as an exploratory control. The publication benchmark instead uses the fixed $+I+G4$ specification. Fixed-model GTR+F+G4 and GTR+F+I+G4 comparisons are reported separately.

All $4^4 = 256$ site-pattern probabilities are enumerated exactly under the correct finite-branch distribution and under the infinite-branch null in Eq. (8). Each simulated alignment is a multinomial draw from one of these exact distributions. The same pattern counts are evaluated by the counterfactual and refined statistics, producing paired decisions without simulator-stream variation.

Alignment lengths are

$$n \in \{100, 500, 1000, 5000, 10000\}. \tag{13}$$

The dense focal-branch grid is

$$t \in \{0.5, 1, 2, 3, 4, 5, 7.5, 10, 12.5, 15, 17.5, 20, 22.5, 25, \\ 30, 35, 40, 45, 50, 60, 75, 100, 150, 200, 500, 1000, 10000\}. \tag{14}$$

The design combines the alignment sizes in Eq. (13) with the focal-branch values in Eq. (14). The branch grid uses 2.5-unit resolution from 10 to 25 and 5-unit resolution from 30 to 50, where the detection transitions occur, while retaining short and extreme controls. For every model, branch length and alignment length, 20,000 null and 20,000 finite-branch alternative alignments are drawn. Half of the null replicates train a formula-specific empirical 95th-percentile threshold; the other half evaluates held-out false-positive control. Power is measured on the independent alternative sample. Wilson intervals quantify individual rejection fractions. Paired percentile-bootstrap intervals use 4,000 resamples of the three discordance outcomes: refined only, agreement and counterfactual only.

Biological sliding-window analyses

The protein 2D and rRNA 3D examples use the alignments and topologies of Hug and colleagues [12]. Branch lengths are re-estimated in IQ-TREE under LG+G4 for the 2,596-site, 3,083-taxon protein alignment and GTR+G4 for the 1,947-site, 1,871-taxon rRNA alignment. Each site is assigned to the rate category with largest posterior probability. Whole-category branch tests use the taxon-pair Bonferroni threshold because a category contains roughly one quarter of the alignment.

Sliding windows contain 36 sites, the minimum scale used in the original analysis. A global variance estimate is formed from the category-specific estimates weighted by their full-alignment site fractions; each window uses its mean coherence and this global variance. The original analysis does not apply Bonferroni correction to windows because one window contains less than 2% of the alignment used to infer the tree. Saturated windows satisfy $z \leq z_\alpha$. Hypervariable rRNA regions V1–V9 and conserved regions C1–C9 are mapped from bacterial 16S annotations after accounting for alignment gaps [15, 19].

The curated rRNA rerun compares published SatuTe with native recomputations of the published and eigenvalue-weighted SatuTe statistics for the Eukaryota and yeast branches. The independent rerun reconstructs site scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both Tree-of-Life datasets. Cross-implementation count differences are reported descriptively and are not treated as paired effects of eigenvalue weighting.

For the topology-rearrangement analysis, the Eukaryota subtree is pruned from the protein 2D tree and regrafted onto the Bacteria–Archaea branch to create a 3D alternative; the reverse operation creates a 2D alternative from the rRNA 3D tree. Branch lengths are re-estimated under the corresponding LG+G4 or GTR+G4 model. Protein-specific scores are calculated for each of 16 concatenated proteins, and rRNA-specific scores for nine annotated regions.

Differences are defined consistently between 2D and 3D placements and compared descriptively, with a two-sample Kolmogorov–Smirnov test for the two empirical difference distributions.

For the HCV example, complete accession-versioned references were selected for subtypes 1a, 1b, 2a, 2b, 3a, 3b, 4a, 5a, 6a and 7a. Polyprotein translations were aligned with MAFFT 7.526 and back-translated; H77 mature-peptide annotations defined Core–NS5B. A full-coding topology was inferred under GTR+F+G4, and the subtype pairs for genotypes 1–3 defined target branches. Codon positions 1+2 and position 3 were concatenated separately. Fixed-topology 36-site windows advanced by 12 sites, with model parameters and branch lengths re-estimated per window. Positions 1+2 and position 3 therefore had equal statistical sample size but spanned 18 and 36 aligned codons.

For every HCV window we recorded the standardized published and eigenvalue-weighted SatuTe statistics and plotted them at the corresponding H77 polyprotein midpoint. The horizontal decision line is $z_{0.01} = 2.326$; a window is counted as saturated when its z-score falls below this line. The 507 positions-1+2 windows and 252 position-3 windows were required for each of three target branches and both statistics, yielding a prespecified 4,554-row grid. We did not correct across overlapping windows because these curves are descriptive localization profiles. The ten-reference design tests partition localization, not population diversity, recombination or genotype assignment.

Data availability

Source data, processed aggregate tables and figure-source files will be deposited in a versioned public archive before publication. The Tree-of-Life alignments are available from the studies in which they were published [12, 13]. HCV records are available from NCBI Nucleotide under accession versions NC_004102.1, AJ238799.1, AB047639.1, D10988.1, D17763.1, D49374.1, Y11604.1, AF064490.1, Y12083.1 and EF108306.2; the accession bundle, alignments, complete 4,554-row window grid and figures are retained in the versioned HCV release. Raw simulation shards are retained on the Vienna Life Science Compute Cluster and are available from the corresponding author on reasonable request. The archival DOI and a reviewer-access link will be added before submission.

Code availability

The development repository is hosted at `https://github.com/EnesSakalliUniWien/iqtree3`. The IQ-TREE integration, simulation and postprocessing scripts, plotting code and manuscript sources used for this study will be archived with the source data before submission. Exact source versions are recorded in the experiment and run manifests.

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Acknowledgements

This work builds on the SatuTe framework of Manuel and colleagues and on its integration with IQ-TREE. Computation used the Vienna Life Science Compute Cluster.

Funding

Funding information will be completed before submission.

Author contributions

B.S. conceived the extension, developed the statistical formulation, performed the analyses and wrote the manuscript.

Competing interests

The author declares no competing interests.

Additional information

Further information on research design is available in the Nature Portfolio Reporting Summary associated with this article.

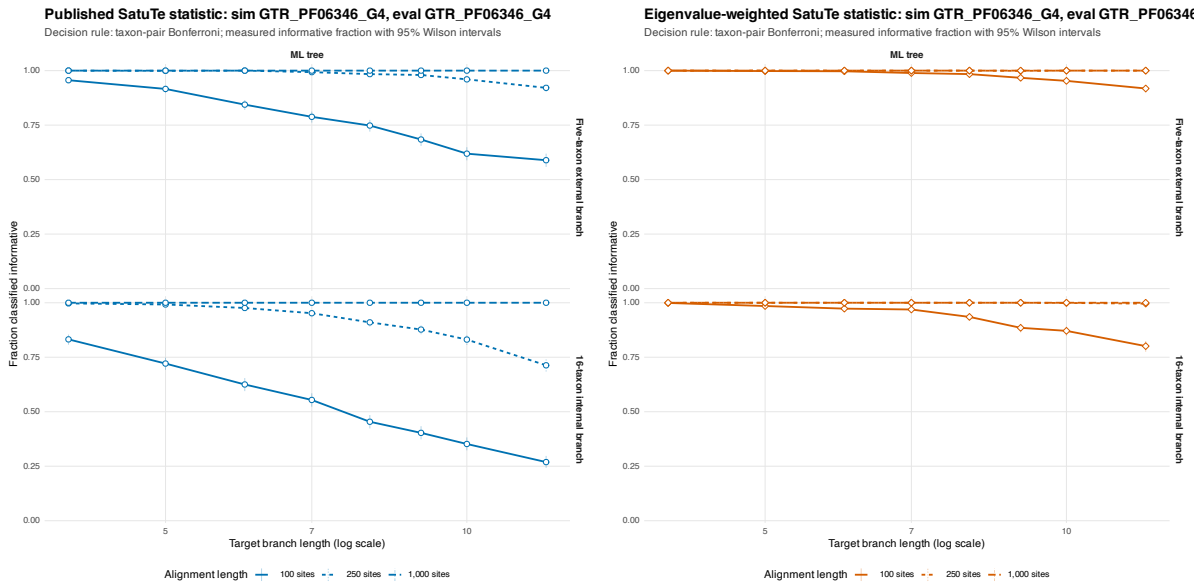


Figure 4: **Matched GTR+F+G4 power. a,** Published SatuTe statistic. **b,** Eigenvalue-weighted SatuTe statistic. Points show fractions of phylogenetically informative alignments among 1,000 paired replicates under taxon Bonferroni; curves are stratified by tree and analysis scenario, and line styles encode alignment length. Eigenvalue weighting preserves power further into the long-branch regime, especially at 100 sites.

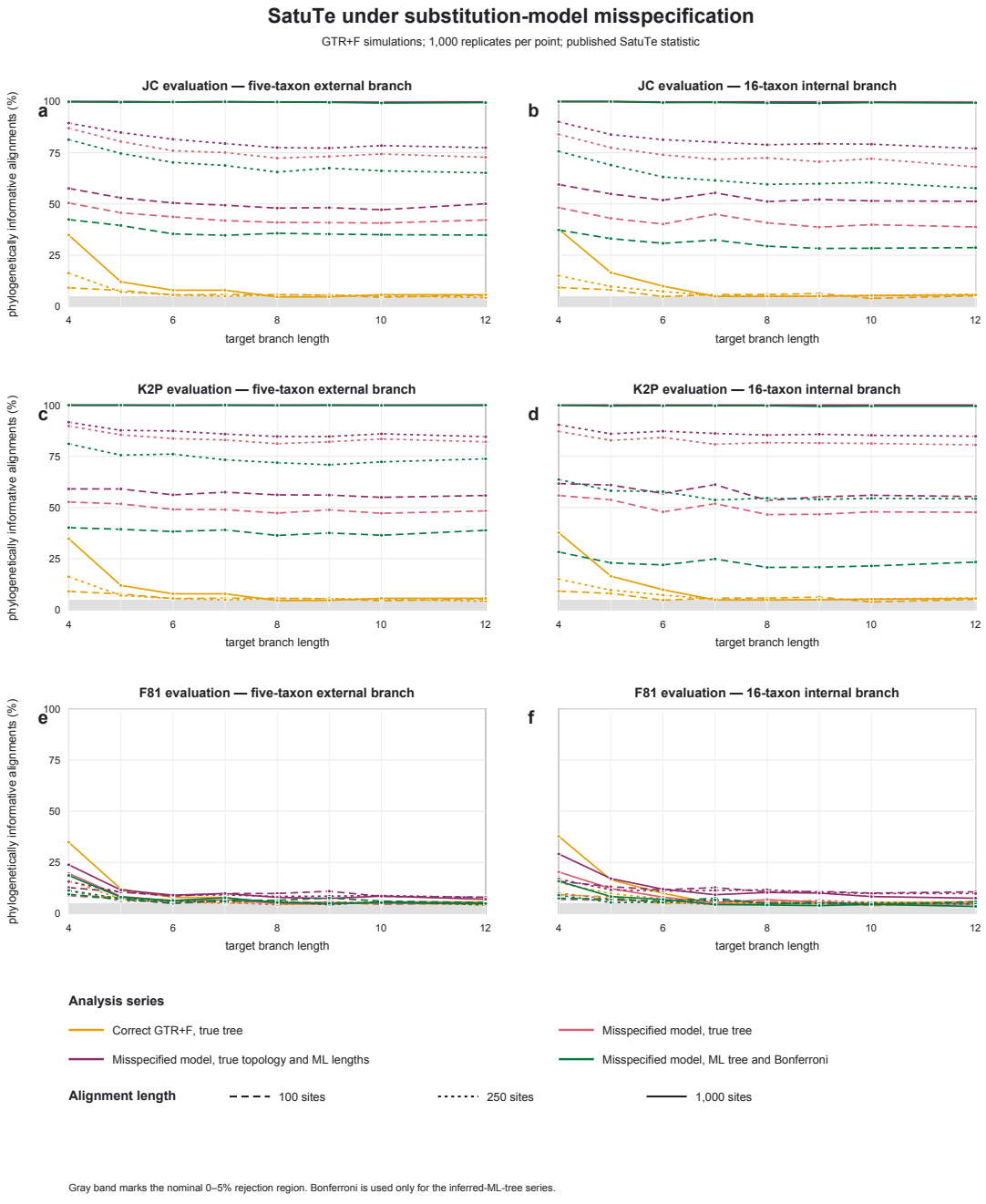


Figure 5: **Model misspecification changes absolute rejection rates.** GTR+F simulations evaluated with JC (**a,b**), K2P (**c,d**) or F81 (**e,f**) on the five-taxon external target (**a,c,e**) and 16-taxon internal target (**b,d,f**). Each panel overlays the correctly specified GTR+F true-tree reference, misspecified true-tree and true-topology/ML-length analyses using unadjusted tests, and the misspecified inferred-ML-tree analysis using taxon-pair Bonferroni. Line style denotes alignment length. JC and K2P produce high spurious rejection at long branches, and this distortion increases rather than disappears with alignment length; F81 is much closer to the correctly specified reference. The gray band marks 0–5%.

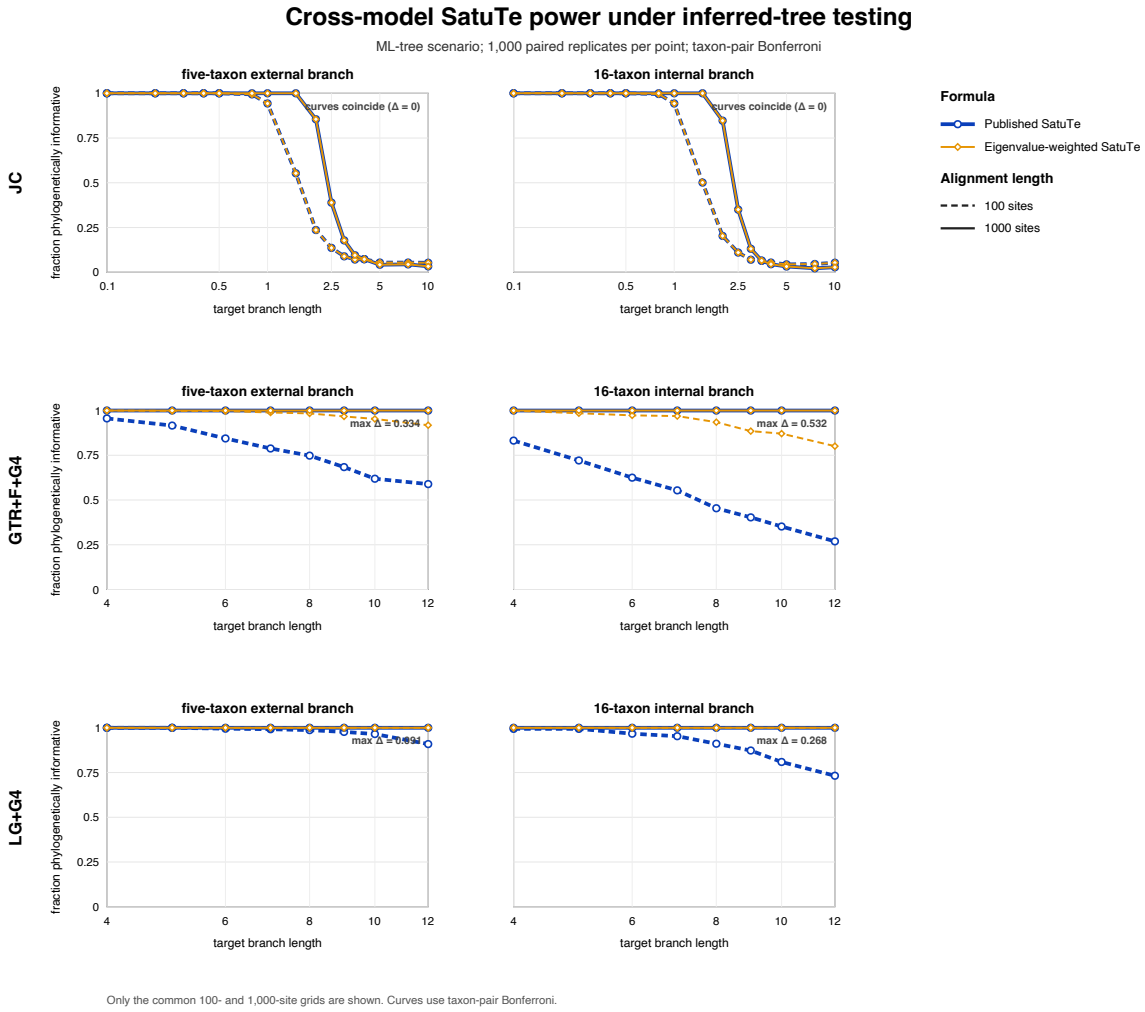


Figure 6: **Eigenvalue weighting extends the detectable-signal range under nucleotide and protein models.** Fractions of phylogenetically informative alignments for the published and eigenvalue-weighted SatuTe statistics in the inferred-ML-tree scenario under taxon-pair Bonferroni, with 1,000 paired replicates per point. Columns show the five-taxon external and 16-taxon internal targets; line style distinguishes the common 100- and 1,000-site grids. Under JC, equal non-stationary eigenvalues make the formulas coincide exactly. GTR+F+G4 shows the largest separation, whereas the LG+G4 gain is smaller and concentrated at the long-branch boundary, particularly for 100-site alignments.

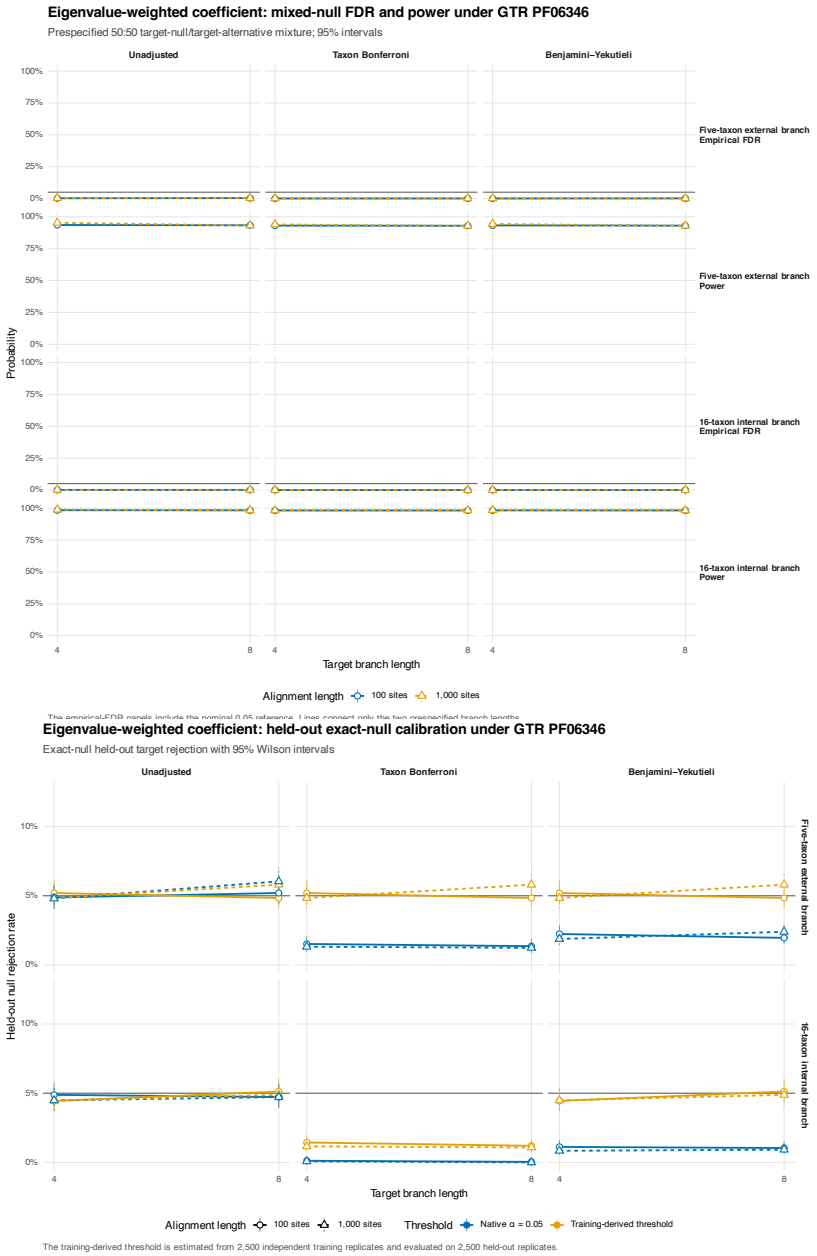


Figure 7: **Independent truth-labelled calibration preserves error control. a,** Empirical FDR and power of the weighted statistic in a prespecified 50:50 target-null/target-alternative GTR mixture. **b,** Native and independently trained thresholds evaluated on 2,500 held-out exact-null replicates per cell. Columns show unadjusted, taxon-Bonferroni and Benjamini–Yekutieli rules; intervals are 95% confidence intervals.

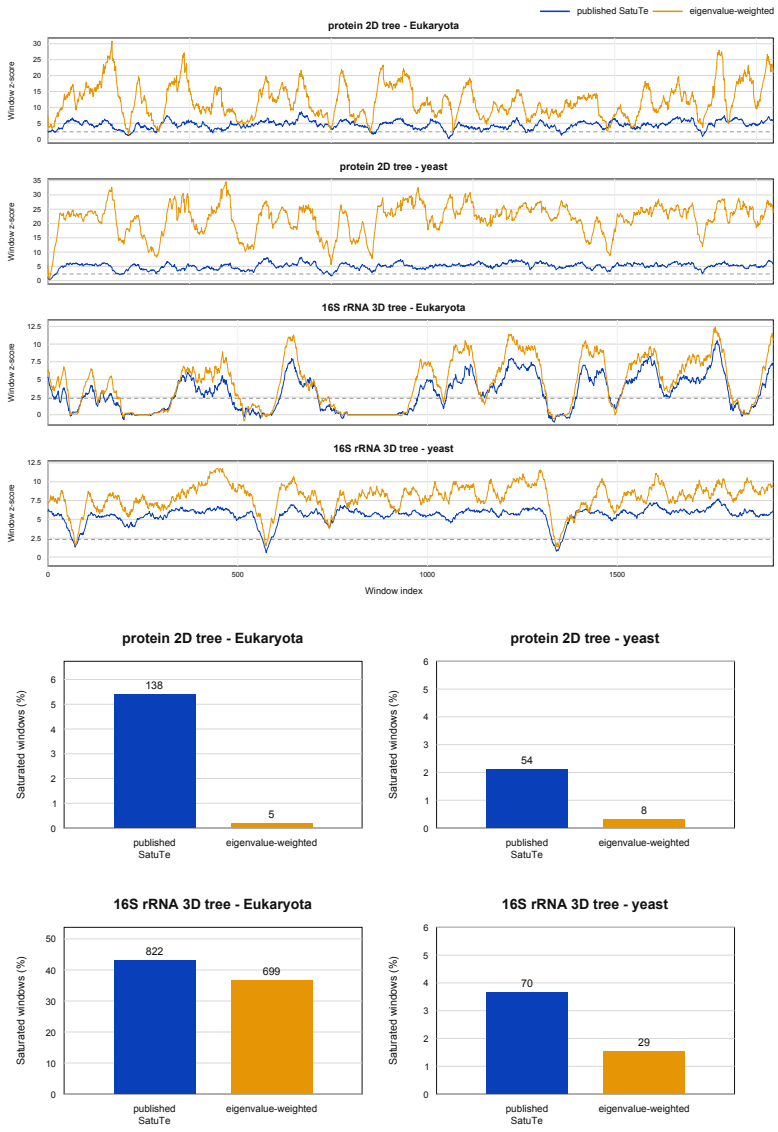


Figure 8: **Branch-specific signal across Tree-of-Life alignments. a,** Thirty-six-site window z-scores for Eukaryota and yeast branches in protein 2D and rRNA 3D trees; the dashed line is the decision threshold. **b,** Fractions and counts of saturated windows. Eigenvalue weighting changes threshold crossings most for the rRNA Eukaryota branch but preserves the protein–rRNA contrast.

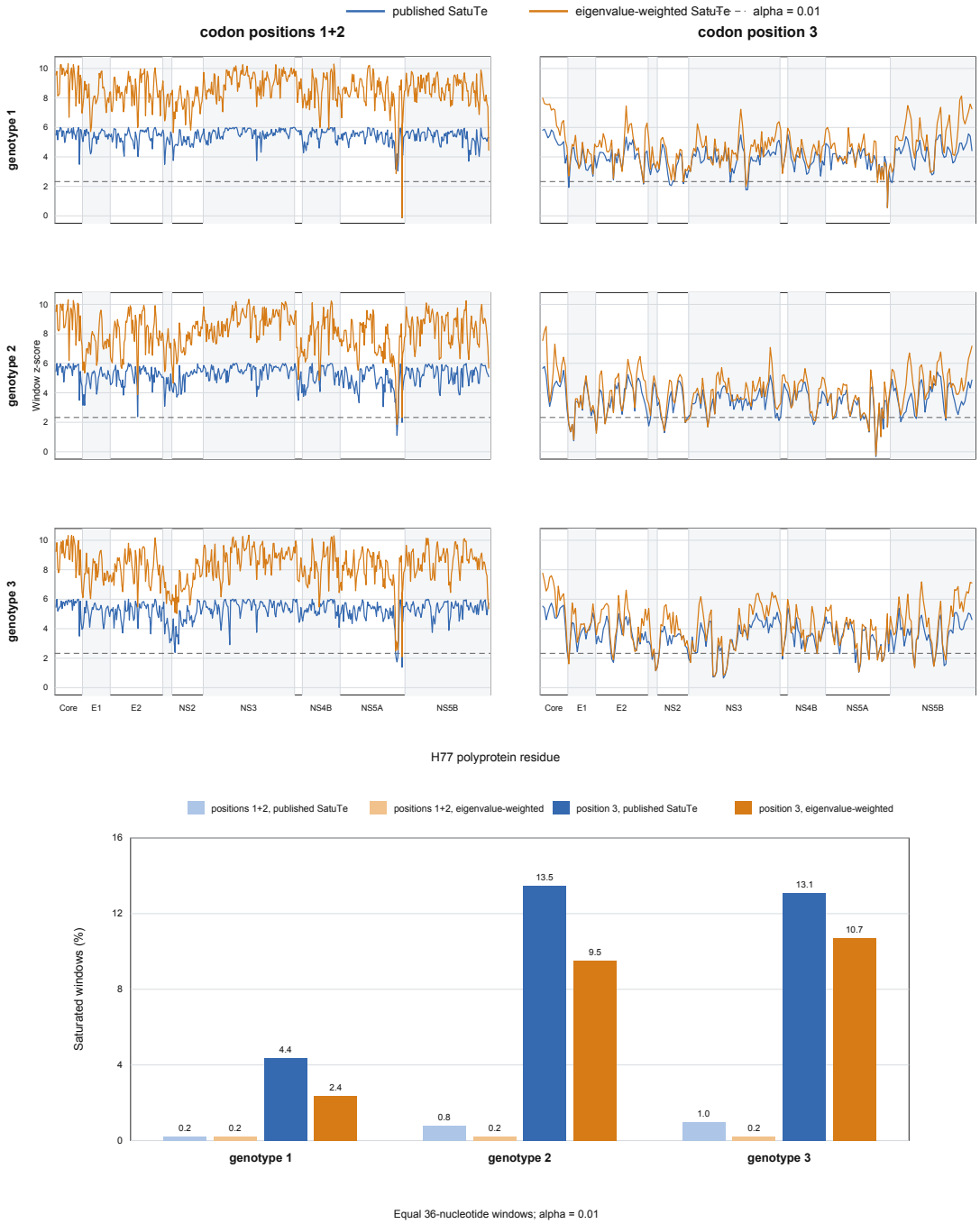


Figure 9: **Codon-position localization in an HCV reference panel. a,** Published and eigenvalue-weighted SatuTe z-scores along the H77 polyprotein coordinate for fixed-topology 36-site windows. The dashed line is the descriptive $\alpha = 0.01$ threshold. **b,** Fractions of saturated windows. Losses concentrate in the synonymous-enriched third-position partition; eigenvalue weighting reduces, but does not remove, a pattern that whole-gene tests conceal.